



One Year Full Membrane Module Warranty

This schedule sets out the warranties with respect to TORAY Membrane Modules. NO OTHER WARRANTIES, EXPRESSED OR IMPLIED, ARE GIVEN EXCEPT AS EXPRESSLY PROVIDED BELOW. THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR PARTICULAR PURPOSE ARE NOT PROVIDED AND ARE EXPRESSLY DISCLAIMED AND EXCLUDED. TORAY IS NOT LIABLE FOR CONSEQUENTIAL, INCIDENTAL, SPECIAL EXEMPLARY AND PUNITIVE DAMAGES, LOSS OF PROFITS, PLANT SHUT DOWN TIME OR SUITS BY THIRD PARTIES AGAINST BUYER, DUE TO A PERFORMANCE FAILURE OR DEFECT OF THE MODULE(S). ALL LIMITATIONS ON LIABILITY SHALL SURVIVE THE EXPIRATION, TERMINATION OR CANCELLATION OF THIS LIMITED WARRANTY. FAILURE OR REFUSAL TO FULLY DISCLOSE THE USE AND OPERATING PARAMETERS OF THE MODULES TO TORAY SHALL RENDER ALL WARRANTIES, OTHER THAN MATERIALS AND WORKMANSHIP, NULL AND VOID.

1. Models

All models of TORAY Membrane Modules specified in TORAY Membrane Module Specification Sheet, except RO and NF membrane products.

2. Definitions

2.1. TORAY

TORAY means Toray Industries, Inc. a corporation duly organized and existing under the laws of Japan and having its principal place of business at 2-1, Nihonbashi-Muromachi 2-chome, Chuo-ku, Tokyo, 103-8666, Japan

2.2. Buyer, Project Reference & Term

“Buyer” - means a company who buys TORAY Membrane Modules directly from TORAY or through TORAY's subsidiaries or through the trading company which TORAY designates.

“Project Reference” – means the information for the project including the location, the design conditions such as feed water quality, required treatment capacity and treated water quality and, operating conditions.

“Term” - means twelve (12) months following the Warranty Start, subject to certain expiry conditions for failure of the Buyer to comply with certain terms and conditions set out in Section 4.

2.3. Membrane Module Replacement Price (hereinafter referred to as “MMRP”)

Unless otherwise stated, the Membrane Module Replacement Price shall be the price taken from TORAY's published retail price list that is in effect on the Module Failure Date for the membrane module(s) required to rectify the Module Failure. In circumstances where a module replacement price has been specified in writing by TORAY in advance, the Membrane Module Replacement Price shall be based on that written specified price, plus the rate of inflation compounded monthly based on the U.S. Consumer Price Index as calculated by the U.S. Bureau of Labor Statistics, unless another inflation index has been specified in the Contract of Sale. All replacement membrane modules are shipped EXW Poway, CA, USA. All freight and any applicable taxes, import duties, brokerage, etc. are the responsibility of the Buyer.

2.4. Module Failure

Module Failure means a situation where, as a result of faulty materials or workmanship, the membrane modules fail to produce permeate which meets the water quality specifications described in the Official Quotation issued by TORAY.

2.5. Module Failure Date

The Module Failure Date shall be the date that written notification of the alleged module failures is received by TORAY as described in Section 4.5.



2.6. Warranty Start Date

For purposes of warranty pro-rating and the effective date for claims commencement, the herein warranty period shall commence on the first date of the membrane modules are placed in operation or six (6) months after shipment of the membrane module(s) from TORAY's facility whichever comes earlier.

3. Warranties and Liabilities

3.1. Materials and Workmanship

TORAY warrants that its membrane module(s) will be free of mechanical defects, due to faulty materials or errors in manufacturing workmanship, which cause the system in which the membrane modules are installed to fail to produce permeate which meets the water quality specifications, described in the Official Quotation issued by TORAY. All claims under the herein warranty must specify the alleged defect in materials or construction or manufacturing workmanship giving rise to such claim as well as the serial number of the relevant membrane module(s). All claims must be submitted to TORAY in writing prior to the expiry of the Term.

3.1.1. TORAY Liability

In full satisfaction of any valid claim hereunder, TORAY, at its sole discretion, shall repair or supply the replacement module for any membrane module that is found by TORAY to be mechanically defective due to faulty material or errors in manufacturing workmanship.

3.1.2. Satisfaction of Claims

TORAY shall have the right to satisfy valid claims under this warranty in a flexible manner in order to restore performance to contractually specified levels of permeate quality. Such flexibility may include the retrofit of newer membrane components or by upgrading failed membrane modules with newer membrane module(s) that embody design and efficiency improvements. Buyer acknowledges that, by virtue of ongoing advances in TORAY's technology and design, fewer compatible replacement membrane modules may, in the future, be required in a given installation to meet contractually specified permeate quality levels. Provided that all contractually specified permeate quality levels are met, should TORAY so proscribe, Buyer consents to the replacement of membrane modules pursuant to this Warranty with a compatible embodiment of TORAY's membrane module technology.

3.2. Price for TORAY Membrane Modules to Replace Modules Originally Shipped with the System

TORAY will replace the originally shipped membrane modules free of charge should Module Failure occurs within the first 12 months from the Warranty Start Date; however, the total number of module(s) to be supplied for free replacement during this period will be less than or equal to the number of initial modules purchased.

Thereafter, the price that the Buyer will pay for TORAY membrane modules supplied under the terms of this Warranty to replace modules originally shipped as part of the Project Reference system shall be calculated as follows:

$$\text{Price} = \frac{\text{Number of whole months elapsed between Module Failure Date – Warranty Start Date}}{\text{Term}} \times (\text{MMRP})$$

3.3. Price for Modules to Replace Replacement TORAY Membrane Modules During the Term

The price that the Buyer will pay for TORAY membrane modules supplied under the terms of this warranty to replace replacement TORAY membrane modules, being modules already supplied under Section 3.2., shall be calculated as follows:

$$\text{Price} = \frac{\text{Number of whole months elapsed between Previous Module Failure Date – New Failure Date}}{\text{Term}} \times (\text{MMRP})$$

3.4. Maximum Term of Warranty

The maximum term of warranty of any replacement module shall not exceed the Term of this Warranty.



4. Conditions of Warranty

4.1. Limitations of Warranties

Occurrence of any of the following as determined by TORAY shall void all warranties hereunder:

- 4.1.1. Physical abuse or misuse of membrane modules;
- 4.1.2. Faulty installation of membrane modules;
- 4.1.3. Unauthorized alteration of any parts originally supplied by TORAY with membrane modules;
- 4.1.4. Failure to strictly and exclusively adhere to TORAY specified membrane module cleaning procedures, including the use of anything other than TORAY-approved membrane module cleaning agents;
- 4.1.5. Failure by the Buyer to follow contractually specified maintenance procedures and schedule;
- 4.1.6. Failure to adhere to contractually defined feed water specifications at all times;
- 4.1.7. Failure to install and maintain a contractually defined agreed TORAY remote data connection to monitor overall plant and membrane module status;
- 4.1.8. Failure to strictly adhere to a TORAY designed or approved maintenance program as updated by TORAY from time to time from the Warranty Start Date;
- 4.1.9. Failure to maintain complete and accurate operating data at all times as described in Section 4.5. below.

4.2. Standard Guidelines

The Buyer shall provide TORAY with the Project Reference beforehand in order for TORAY to issue the Official Quotation including permeate water quality specifications. This Warranty is conditional upon the shipping, storage, system design, installation, operation and maintenance of the membrane modules in strict accordance with the TORAY membrane module operations and maintenance guidelines as updated by TORAY from time to time. Buyer authorizes TORAY to conduct any reasonable review of system design or to inspect facilities where membrane modules are installed upon reasonable notice to the Buyer. Such reviews and/or inspections are intended to assist TORAY and the Buyer in detection of system faults and to optimize the care and operation of the membrane modules.

4.3. Additional Equipment

TORAY assumes no liability for any additional equipment not authorized by TORAY that may be installed in or connected to the plant after operation has begun.

4.4. Warranty Maintenance

To maintain the warranty described herein, plant operating records from initial start-up date until claim must be maintained and made available to TORAY upon request. Such documentation must be provided in detail as specified in 4.5 below in order to: a) verify uninterrupted compliance with guidelines; and b) establish liability for module(s) replaced or repaired under warranty. **IT IS AGREED AND UNDERSTOOD THAT THE BUYER SHALL NOT BE ENTITLED TO MAKE ANY CLAIMS UNDER THIS WARRANTY IF THE MEMBRANE MODULES HAVE NOT BEEN OPERATED IN STRICT ACCORDANCE WITH THE TORAY MEMBRANE MODULE OPERATIONS AND MAINTENANCE GUIDELINES AS UPDATED BY TORAY FROM TIME TO TIME.** Buyer must measure and maintain records to establish that the membrane module(s) have been operated in accordance with such guidelines, failing which all warranties and rights of the customer shall be null and void.

4.5. Notification of Performance Deficiency

All claims filed hereunder shall be made in writing within thirty (30) days of identifying a defect in materials or workmanship and shall present a detailed analysis of the system and individual module data showing the performance deficiency, and must include: The serial number(s) of the module(s) involved, and the individual membrane module(s) operating data, system operating data or defect in materials or workmanship upon which the claim is based. Operating data must include regular information on: flow, trans-membrane pressure, contractually specified feed water quality parameters and temperature, and elapsed time since start-up (days).

TORAY reserves the right to require additional data as necessary to validate claims filed. The Module Failure Date shall be the date that written notification of the alleged module failure(s) is received by TORAY.

4.6. Verification of Performance Deficiency

After receipt of notice, TORAY will promptly undertake such investigations as, in TORAY's opinion are necessary to verify whether a deficiency exists and to establish liability for remedy of any deficiency. Buyer may, in course of these investigations, be requested to return module(s) to TORAY for examination. TORAY may also conduct reasonable tests and inspections that are standard in the industry (hereinafter referred to as "Tests") at end-user's or at Buyer's premises. If, following the performance of such Tests, it is determined that the membrane modules are not defective and/or there is a problem or deficiency whose remedy is not the responsibility of TORAY, Buyer shall reimburse TORAY for all reasonable costs and expenses associated with said inspections and Tests, including applicable travel expenses.

4.7. Return Procedure

In the event that the return of a membrane module(s) is required pursuant to this Warranty, Buyer shall first obtain a Return Goods Authorization (hereinafter referred to as "RGA") number from TORAY. Membrane module(s) shipped to TORAY for warranty examination must be shipped freight prepaid. If Buyer desires temporary replacement membrane module(s) to replace those alleged to be defective and returned to TORAY for warranty examination, Buyer is responsible for the cost associated with any such replacements until examination of the returned membrane modules pursuant to this Warranty is complete. Membrane module(s) examined as part of a warranty claim which are subsequently found by TORAY to be performing as warranted will be returned to Buyer, freight collect.



4.8. Required Data Logging Items:

Analog values for:

- Permeate pressure during filtration
- Permeate pressure during relax
- Permeate flow during filtration
- Liquid temperature in MBR tank
- Mixed liquor feed (or return) flow
- Scouring air flow
- Level in MBR tank
- Excess sludge flow

Digital counts for:

- Permeate quantity during filtration
- Excess sludge quantity

Timer settings for:

- Filtration time / relax time
- Air diffuser cleaning interval

On-line and/or laboratory results for:

- MLSS in mixed liquor
- pH in mixed liquor
- Turbidity in permeate
- COD in permeate
- NH₄-N in permeate

Calculated results for:

- Trans membrane pressure (P static-P suction)
- Permeate flux rate (Perm. Flow l/h / area m²)

Operator remarks:

- Area for manual input of remarks from the operator